

Orca Bricks — Research & Ideas Sheets

FLL BIOGLOW 2026-27 Innovation Project · typed up October 4, 2026

Seven Orca Bricks team members filled in a Research & Ideas Sheet for the FLL BIOGLOW 2026-27 Innovation Project, most dated September 27, 2026. Every sheet names the same problem: deforestation, water pollution and air pollution. This is a typed copy of the handwritten sheets, with spelling cleaned up and uncertain words marked (?).

Ideas at a glance

#	TEAM MEMBER	BIG IDEA
1	Lydia	A huge machine that plants native seeds; a traffic-light water pollution monitor with a filter
2	Chansy J.	Spread the word about pollution, person to person
3	Krishiv	Make little homes for animals
4	Iraj	A water monitor that triggers a purifier when toxic levels are too high
5	Ishaani	A robot trash can that comes to you when you need it
6	Shaurya	Switch to natural gases as a different type of fuel
7	Devansh Molagavalli	A simple model showing how to help solve all three problems

Lydia and Iraj both proposed a water monitor paired with a filter or purifier.

1 Lydia *Dated Sep. 27, 2026. Two pages, with a different idea on each side.*

The problem we care about

Deforestation, water and air pollution.

Who does it affect? Why does it matter?

It affects animals and plants, and that can impact the ecosystem by taking away natural habitats.

What we found out

(1) There was a project to plant 1 trillion trees. (2) In South Korea they are doing reforestation. (3) If you plant the same trees to form an ecosystem, it won't work.

Idea 1 — Native seed planter (front)

Big idea: A huge seed planter machine that deposits native seeds. **Sketch:** A wheeled machine with a wide row of planting teeth across the front. **How it works:** It plants different native seeds into the ground to form an ecosystem. **Build to test:** A lot of soil without any seeds. **Share:** Social media. **Questions:** Left blank.

Idea 2 — Water pollution monitor (back)

Big idea: A monitor that can tell people if the water is too polluted. **Sketch:** A traffic light. **How it works:** Like a traffic light — red when too polluted, yellow when a little polluted, green when clear. If polluted, the water goes through a filter to filter out the germs. **Build to test:** A big bucket of water to pollute. **Share:** Social media.

2 Chansy J. Dated "September". Spelled Chansy as it reads on the sheet; spelling to be confirmed.

The problem we care about

Air pollution, water pollution, deforestation all kill animals. How does it impact biodiversity?

Who does it affect? Why does it matter?

It affects wildlife and us. It matters because if we don't act upon it the population will die.

What we found out

(1) Majority of ecosystem is underground. (2) Pollution is getting into animals. (3) Air pollution is bad for our lungs.

Our big idea

Tell everyone how air pollution affects us and why it's important — and tell them to tell everyone they know.

Sketch: Two stick figures, one talking to the other. **How it works:** After we tell our friends, they'll tell people to tell their friends, and so on. **Build to test:** "Nothing 2 build." **Share:** Word of mouth, texting. **Questions:** What if no one cares? What if word stops and everyone forgets?

3 Krishiv No date on the sheet.

The problem we care about

Deforestation, water pollution, air pollution.

Who does it affect? Why does it matter?

Animals, us.

What we found out

(1) 3,000 animals die. (2) More animals are expected to die. (3) It affects us too.

Our big idea

Make little homes. **Sketch:** A stick figure next to a small house. **How it works:** It is a small home. **Build to test:** Home. **Share:** Go to a zoo. **Questions:** How much money?

4 Iraj Dated 9/27.

The problem we care about

Deforestation, water pollution, and air pollution.

Who does it affect? Why does it matter?

It affects wildlife such as animals, plants, and trees. It matters because if we did not stop, we would end up getting ourselves extinct, because there will be no food or oxygen left for us.

What we found out

(1) Almost 30% of animals on Earth are endangered or basically extinct. (2) Some places on Earth are preserved for animals. (3) "CSAs (?) are helpers": growing multiple crops helps soil thrive (first word hard to read).

Our big idea

The monitor will go off if the toxic level is too high, and the purifier will clean the water. **Sketch:** Dirty water flowing into a water purifier with clean water below, and a monitor at the side. **How it works:** When the monitor goes off, the water flows through the purifier and gets cleaned for drinking. **Build to test:** A Lego pool with a retractable hole. **Share:** Raise awareness that we are getting closer to extinction if we pollute water and air and destroy forests. **Questions:** How to clean air.

5 Ishaani *Dated Sep. 27. Two pages: the sheet and a back page with more research detail.*

The problem we care about

Deforestation, water pollution, and air pollution.

Who does it affect? Why does it matter?

If we never had forests, water or air, none of the humans, animals, and plants would survive.

What we found out

(1) There are species we don't know yet that are already going extinct. (2) After people started making new inventions, biodiversity kept going down. (3) People have tried to help biodiversity, but every year it gets worse.

More on the research (back page)

Point 2: New plant-based inventions like gases and new fertilizers can harm habitats and animals. **Point 3:** People have cleaned up lakes, beaches and rivers and tried to stop using pesticides, but they get dirty again and pesticides are still used over compost.

Our big idea

Build a robot that knows when someone needs a trash can and comes so the person can throw the trash away. We could also walk, bike, or scooter instead of cars. **Sketch:** People outdoors near trees, with a robot trash can on wheels rolling up to them. **How it works:** When someone needs to throw something away, the robot comes to them. **Build to test:** Put different pieces of trash around the beach to see if it picks them up. **Share:** Unfinished (only "Sh" written). **Questions:** If we create AI robots, won't they destroy nature while trying to help us?

6 Shaurya *No name on the sheet. Dated 6/27/26, probably meant as 9/27/26.*

The problem we care about

Deforestation, air pollution, and water pollution.

Who does it affect? Why does it matter?

It affects the animals that will eventually go extinct.

What we found out

Left blank.

Our big idea

Switch to natural gases. **Sketch:** A truck at a gas pump, crossed out, next to a truck at a different kind of station.

How it works: Use a different type of fuel. **Build to test:** Test different fuels. **Share:** Talk to politicians.

Questions: Will it work?

7 Devansh Molagavalli *No date on the sheet.*

The problem we care about

Air pollution, deforestation and water pollution.

Who does it affect? Why does it matter?

It means no trees and bad air pollution, and the sea animals will die.

What we found out

(1) Air pollution can harm plants, animals and insects, damage forests, and contribute to climate change. (2) Animals lose their homes, plants disappear, and some species may become endangered. (3) Fish, turtles, birds and other aquatic organisms can be injured, poisoned... (writing runs off the page).

Our big idea

Create a simple model showing how we can help solve three major environmental problems: air pollution, deforestation and water pollution. **Sketch:** A sun-style diagram with a circle in the middle and labeled rays: save animals, grow trees, save energy, don't cut trees, save water, recycle, stop pollution, don't litter. **How it works:** Faint and erased, possibly "build a bird house" (?). **Build to test / share / questions:** Left blank.